



Two of Vista's key features will make for a "sexy time for motherboards": ReadyBoost and SideShow

Stop us if you've heard this one—"talking about my computer makes my mother bored" (snigger). Ahem—no, not funny. Motherboards are getting a bit dull, though. We remember the good old days when we would break out into fist-fights in our *Digit* cantina over the relative merits of an Intel chipset versus an NVIDIA, ATI, or VIA offering; on certain moonless nights, even SiS products were whispered into the darkness. Ah yes, those were the days—today, though, market consolidations and standardisations have taken their toll on our most beloved of simple pleasures. Almost anyone would agree that a good chipset must either be an Intel or an NVIDIA—where's the fun? All you have left to differentiate one motherboard from another is the colour of its PCB ("Look, a RED one!"). Very boring!

Respite might come from an unlikely source, though: Windows Vista! Vista might not be the consumer's best friend, but it is certainly a great window of opportunity for hardware vendors. As Vista goes, two of its key features will make for a much needed "sexy time for motherboards": ReadyBoost and SideShow.

ReadyBoost can use storage space on some removable media devices, such as USB Flash drives, to speed up your computer. Thanks to unprecedented demand and even greater supply, Flash memory is cheap and probably sold by

Motherboards

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the kilo in some backstreets in China. You should soon see motherboards with embedded Flash storage to leverage this Vista feature.

What the Flash would help with would be caching of frequently-used programs. Vista has a nifty feature called SuperFetch, through which it monitors stuff you run often and caches them to memory, so that future instances of your favourite programs start quicker. Another neat trick that a motherboard with embedded Flash memory might pull off would be to act as an overgrown floppy drive: imagine booting into your favourite Linux distribution, or even a DOS variant upon startup.

SideShow is another neat feature that Vista supports—a SideShow device is like a mini-monitor that displays key information from your Vista system. It could either be a small LCD window on the lid of a laptop which shows you unread mails, battery life and so on when the unit is off, or it might be a clock that shows you your IM buddies, your currently-playing track list or the weather even when your monitor is switched off. For a PC, SideShow would make a great addition if it is wireless—imagine IM and mail notifications while lounging in your living room.

So much for new features—what about chipsets? NVIDIA chipsets are the way to go for AMD processors (perhaps until the time AMD truly assimilates ATI offerings into its product lines). For Intel processors, the roadmap calls for at least one SiS-based motherboard; time will be judge of that product's performance. Today, however, Intel's P965 chipset is the one to watch—it offers product support across the entire Intel line from Pentium D to dual- and even quad-core processors. Barring some performance hiccups (in some

instances, the 975X still performs better), and some bad decisions (such as no native support for IDE drives), the P965 would make a great platform for any future Intel purchase. As for choice, the NVIDIA 600i family is perhaps the best route for Intel gamers to take. While the top-end 680i range came with its baggage of problems—from audio issues to data corruption—the problems are largely fixed now, and it is the best non-Intel solution for the high-end. More interesting in the 600 family is the lower-priced 650i chipset, which is meant as a direct competitor to Intel's P965.

The last few months have seen muted support in terms of chipsets for AMD processors. The reason for this could be two-fold: one, AMD bought ATI, so other chipset makers are unsure of how to react; and two, Intel is once again the performance leader with its Core 2 Duo range of processors. There is less noise in the marketing circles for AMD chipsets.

While on AMD, it would be remiss not to mention the new DTX standard the company has proposed for motherboards. DTX will essentially be an extension of the ATX standard for small-form-factor systems. The hardware spec defines two types of motherboards: a standard DTX board, 9.6-inch x 8-inch, and also a smaller mini-DTX unit at 6.7-inch x 8-inch. DTX systems, though small, would be able to accommodate a processor with a thermal envelope of up to 65 W, as well as two expansion cards which could be either PCI or PCI-Express. The form factor also calls for a notebook-style ExpressCard PC card. AMD has already released a review copy of the new standard, and is encouraging contribution from the market to design and evolve the form factor.

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